

Nitish Ravisankar Raveendran

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College Park, MD

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Education

University of Maryland, College Park

Master of Engineering in Robotics

Expected Graduation: Spring, 2025

Relevant Coursework: Advanced Robotics, Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion

Anna University, PEC, Chennai

Bachelor of Engineering in Electronics and Communications Engineering

July 2022

CGPA: 8.94/10.0

Relevant Experience

Robotics Club, PEC Chennai *President*

March 2019 - August 2022

- Championed club initiatives that increased membership and orchestrated multiple innovative projects and events, engaging a large number of students.

- Enhanced application security by implementing OAuth-based authentication, reducing vulnerabilities.

Institute of Innovations, Chennai *Intern*

May - June 2021

- Designed and deployed IoT devices using ESP8266 and ESP-32, improving real-time data acquisition by 20%.

- Integrated cloud-based analytics, reducing latency in sensor data processing.

Projects

Bipedal Locomotion Using Reinforcement Learning

Fall 2024

- Implemented TRPO, PPO, and TQC to generate adaptive bipedal gaits, reducing gait instability by 40% and cutting training convergence time by 35% to achieve reliable walking in simulation.

Gesture-Based Control in Assistive Technology

Fall 2024

- Developed a real-time hand-gesture interface with Google MediaPipe and ROS2 to enable intuitive robot teleoperation, reducing operator fatigue.

Arduino-Based Aimbot with YOLO Object Detection

Summer 2024

- Developed an Arduino-driven targeting system using YOLOv10, achieving 95% detection accuracy and reducing actuation latency by 50% for rapid, reliable servo adjustments.

Informed Quick-RRT* Path Planning

Spring 2024

- Combined informed ellipsoidal sampling from Informed RRT* with Quick-RRT*'s ancestor-based rewiring to concentrate search on high-utility regions, reducing convergence time and path cost.

Reinforcement Learning for Pursuit-Evasion

Spring 2024

- Built a DQN-based framework in ROS2 & Gazebo to investigate multi-agent pursuit-evasion strategies, training evader and pursuer TurtleBot3 agents to achieve stable capture/escape behaviors

TurtleBot3 Vision-Based Path Following

Fall 2023

- Designed a ROS2+OpenCV perception pipeline to enhance TurtleBot3's autonomous navigation by following predefined waypoints and safely stopping at stop signs or obstacles, increasing route success rates by 40% while maintaining real-time detection at 30 FPS.

ROS2 Path Automation

Fall 2023

- Designed to automate Nav2 goal generation by converting logical camera and ArUco marker detections into 'Navigate-ToPose' actions, eliminating manual waypoint setup and significantly reducing configuration effort.

Alpha - Pick and Place Mobile Manipulator

Fall 2023

- Developed a CAD-based mobile manipulator with ROS2 integration to automate precision pick-and-place tasks with improved reliability.

Quadcopter Design and Development

Summer 2022

- Designed and programmed an autonomous quadcopter with closed-loop PID control in Python and C++, enhancing flight precision.

Technical Skills

Languages & Tools: Python, C++, Bash, Git, Docker, pytest, GitHub Actions

Computer Vision & ML: OpenCV, MediaPipe, YOLO, PyTorch, TensorFlow

Platforms: Ubuntu/Linux, ROS2

SBC: Raspberry Pi, NVIDIA Jetson

Embedded: Arduino, ESP8266/32